Available online at www.sciencedirect.com
SCIENCE DIRECT®
Tectonophysics 377 (2003) 119–141
TECTONOPHYSICS
www.elsevier.com/locate/tecto

Palaeomagnetic study of the Ronda peridotites (Betic Cordillera, southern Spain)

V. Villasante-Marcos[a,*], M.L. Osete[a,1], F. Gervilla[b,2], V. García-Dueñas[c,3]

[a] *Departamento de Física de la Tierra, Astronomía y Astrofísica I, Facultad de Ciencias Físicas, Universidad Complutense de Madrid, Avda. Complutense, s/n, 28040, Madrid, Spain*

[b] *Facultad de Ciencias, Instituto Andaluz de Ciencias de la Tierra, CSIC-Universidad de Granada, Avda. Fuentenueva s/n, 18002, Granada, Spain*

[c] *Facultad de Ciencias, Departamento de Geodinámica, Universidad de Granada, Avda. Fuentenueva s/n, 18002, Granada, Spain*

Received 20 October 2002; received in revised form 10 July 2003; accepted 25 August 2003

Abstract

A palaeomagnetic study of the Ronda peridotites (southern Spain) has been carried out on 301 samples from 20 sites, spread along the three main outcrops of the ultrabasic complex: Ronda, Ojén and Carratraca massifs. Different lithologies and outcrops with different degrees of serpentinization have been sampled and analysed. Rock magnetic experiments have been carried out on a representative set of samples. These measurements include: Curie curves, hysteresis cycles, isothermal remanent magnetization (IRM) acquisition curves, thermal demagnetization of IRM imparted along three orthogonal axes and magnetic bulk susceptibility. Results indicate that magnetite is the main magnetic mineral present in the samples. Stepwise thermal and alternating field (AF) demagnetization of the natural remanent magnetization (NRM) reveals the presence of a characteristic remanent magnetization (ChRM) carried by magnetite, and in some sepentinized samples, a northward component with variable unblocking temperatures up to 250–575 °C. The appearance and the relative intensity of this northward component are strongly related to serpentinization degree. Taking into account the geological history of the peridotites, the ChRM has been considered as a thermo-chemical remanent magnetization acquired during the first serpentinization phase associated to the post-metamorphic cooling of this unit. On the basis of radiometric and fission track analysis, the ChRM is proposed to have been acquired between 20 and 17–18 Ma. The inclination of the mean direction of the ChRM statistically coincides with the expected inclination for stable Iberia during the Oligocene–Miocene. The declination of the ChRM differs from the expected declination, indicating clockwise block rotations of $41 \pm 12^{\circ}$ about vertical axes since the cooling of the peridotites. When applying a compositional layering correction, the ChRM directions fail to pass this kind of fold test, thus, the compositional layering was not a palaeohorizontal during ChRM acquisition time. Normal and reversed polarities of the ChRM are reported, showing that at least one reversal of the Earth's magnetic field took place during ChRM acquisition process. A tentative polarity zonation within the peridotitic outcrops is also suggested.

* Corresponding author. Tel.: +34-91-394-44-40, +34-91-394-43-96.

E-mail addresses: vicvilla@fis.ucm.es (V. Villasante-Marcos), mlosete@fis.ucm.es (M.L. Osete), gervilla@goliat.ugr.es (F. Gervilla), vgarciad@ugr.es (V. García-Dueñas).

[1] Tel.: +34-91-394-43-96.

[2] Tel.: +34-958-24-66-17.

[3] Tel.: +34-958-24-33-50.

0040-1951/$ - see front matter © 2003 Elsevier B.V. All rights reserved.
doi:10.1016/j.tecto.2003.08.023

No evidence is found from these data for the previously proposed simultaneity between post-metamorphic cooling and rotation of the peridotites.
© 2003 Elsevier B.V. All rights reserved.

Keywords: Palaeomagnetism; Ronda peridotites; Block rotations; Betic Cordilleras

1. Introduction

The Betic Cordilleras, in the south of the Iberian Peninsula, is the northern branch of the Betic–Rifean orogen, an arch-shaped mountain belt bordering the Alborán Sea, which constitutes the westernmost segment of the peri-Mediterranean Alpine collisional system. The orogen was formed in the Africa–Iberia borderland during convergence of these two lithospheric plates from Late Cretaceous to Tertiary times.

The Betic–Rifean orogen is divided in four domains (Balanyá and García-Dueñas, 1987; Balanyá, 1991): the SudIberian domain (the External Zones of the Betics), corresponding to the Iberian passive palaeomargin; the Maghrebian domain (the External Zones of the Rif), corresponding to the African passive palaeomargin; the Alborán domain; and the Flysch Trough allochtonous sedimentary units. The Alborán domain is formed by the Alborán basin and the allochtonous nappe complexes that were thrust over the SudIberian and Maghrebian domains and that constitute the Internal Zones of the Betic and Rif Cordilleras. In the Betics, the Internal Zones are made up of several thrust sheets that have been traditionally grouped into three main tectonic complexes. In ascending order, these are: (1) the *Nevado–Filábride complex*; (2) the *Alpujárride complex*; and (3) the *Maláguide complex*. The two first complexes experienced plurifacial metamorphism, mainly of Alpine age, whereas the Maláguide complex has experienced very low metamorphism of probably Hercynian age (Chalouan and Michard, 1990). The main outcrops of ultramafic rocks in the Betic Cordillera, the Ronda peridotites, are located in the western units of the Alpujárride complex. The Rifean related units are the Beni Bousera peridotites, part of the Sebtide nappe complex, which is the Rifean version of the Alpujárride complex.

In order to explain the geometry and the evolution of the Betic–Rifean region, different models have been proposed. They include oroclinal bending of an originally straight convergent mountain chain between Africa and Europe (Carey, 1955), deformation around a west-driving Alborán microplate located between Africa and Europe (Andrieux et al., 1971; Leblanc and Olivier, 1984), outwardly directed thrusting driven by the gravitational collapse of the root of an engrossed collisional orogen (Platt and Vissers, 1989), upwelling of a mantle diapir below the basin (Weijermaars, 1985; Doblas and Oyarzun, 1989), or a delamination process in an asymmetric lithospheric mantle (García-Dueñas et al., 1992; Docherty and Banda, 1995). Palaeomagnetic studies in the Betic–Rif region have demonstrated that important rotations around vertical axis have occurred in this area. Therefore, any proposed geodynamic model should explain the observed block rotations. In addition, rotations have to be taken into account in structural analysis, because fold trends or kinematic indicators could have been rotated.

Several palaeomagnetic studies have discovered a general pattern of clockwise rotations in the External Zones of the Betics (Osete et al., 1988, 1989; Platzman and Lowrie, 1992; Platzman, 1992; Allerton, 1994; Allerton et al., 1993, 1994; Platt et al., 1994), and dominant anticlockwise rotations in the Rif (Platzman et al., 1993). Most of these studies were carried out on Jurassic and Cretaceous sedimentary rocks. Therefore, the timing of the rotations could not be established by these palaeomagnetic data. Villalain et al. (1994, 1996) discovered the existence of a pervasive remagnetization that affected Jurassic limestones of the whole western Subbetic zone. This remagnetization was coeval with the folding of the studied units and was dated as Neogene. Since the remagnetized component was also rotated, it placed the timing of the block rotations as post-Paleogene.

Kirker and McClelland (1996) found similar results in other lithologies of the western Subbetics. Recently, Osete et al. (2004) have extended these studies to Jurassic limestones from the central and eastern Subbetics. They have also found an important remagnetization, of similar characteristics to that reported in the western Subbetics, which is also contemporary with the Neogene folding of this region. But, in contrast to the results in the western Subbetics, where rotations occurred after the remagnetization event, in the central part of the Subbetic Zone, they were completed by the time of the remagnetization process.

In contrast to the attention focused on the External Betics, few studies have dealt with the Internal Zones, in part due to the absence of favourable lithologies. Calvo et al. (1994, 1997) studied Tertiary and Quaternary volcanic and sedimentary rocks from southeastern Iberia, finding heterogeneous rotations in the proximity of important strike-slip faults, and regions that have not experienced rotation at all. Allerton et al. (1993) studied in detail a section in Sierra Espuña (Eastern Internal Zone) where they sampled sediments of upper Miocene, upper Oligocene–lower Miocene, Oligocene, Jurassic and Permo-Triassic ages. Large rotations were found in this region (up to 200°). Results suggested that about 60° of clockwise rotation occurred in the latest Oligocene–earliest Miocene, and a further 140° of clockwise rotation subsequently. Large rotations of about 90–140° have been found affecting a suite of Oligocene–early Miocene mafic dykes intruded into the Maláguide allochthon in the Internal Zone of the Betics (Platzman et al., 2000; Calvo et al., 2001). These rotations were considered to happen during the early Miocene related to thrusting over the External Zones, but no direct evidence could be shown.

Elazzab and Feinberg (1994) studied the serpentinized peridotites of Beni Malek, in the External Zones of the Rif. This was the first palaeomagnetic study carried out on ultrabasic rocks of the Betic–Rifean Cordilleras. They found counter-clockwise rotations of $14 \pm 11°$. Saddiqi et al. (1995) worked on the peridotitic massif of Beni Bousera, in the Internal Zones of the Rif, finding counter-clockwise rotations of $74 \pm 11°$. In the Ronda peridotites, Feinberg et al. (1996) carried out a palaeomagnetic study in 15 sites in the Ronda and Ojen massifs, including peridotites, granites intruding the peridotites and their country rocks. They found that clockwise rotations of $46 \pm 10°$ affected these units. The acquisition of the remanence was dated as Aquitanian–Burdigalian, being attributed to the post-metamorphic cooling of the Alpujárrides. They dated the timing of the rotations on the basis of palaeomagnetic data, considering the existence in some samples of a northward natural remanent magnetization (NRM) low temperature component, with maximum unblocking temperatures of 250–450 °C. They concluded that the rotation of the units was simultaneous with the post-metamorphic cooling. This could be the only independent data to date the rotations of this area. But a northward-directed component could also have been interpreted as a viscous magnetization or a secondary present-day field component.

This study has two goals: (1) to investigate the characteristics and origin of the low temperature northward-directed component of the Ronda peridotites; and (2) to better determine the spatial distribution of the rotations including more data from new outcrops. The study of the timing of rotations and their spatial distribution offers important information on the mechanism responsible of the block rotations.

2. Geologic setting: the Ronda peridotites

The main outcrops of ultramafic rocks in the Betic Cordillera are located in the western units of the Alpujárride complex (Fig. 1). They are distributed in three main massifs (Ronda, with an area ~ 300 km^2; Ojén, ~ 70 km^2; and Carratraca, ~ 60 km^2) with different degrees of serpentinization (the larger the less serpentinized), together with several smaller, highly serpentinized massifs. The ultramafic bodies (1.5–4.5 km width) constitute the lowest part of Los Reales nappe, below a sequence of metasedimentary rocks known as the Casares tectonic unit. The ultramafites, in turn, overthrust the Blanca unit (Fig. 1).

The Ronda peridotites are mainly spinel lherzolites and plagioclase lherzolites, but harzburgites, dunites and some layers of pyroxenites are also

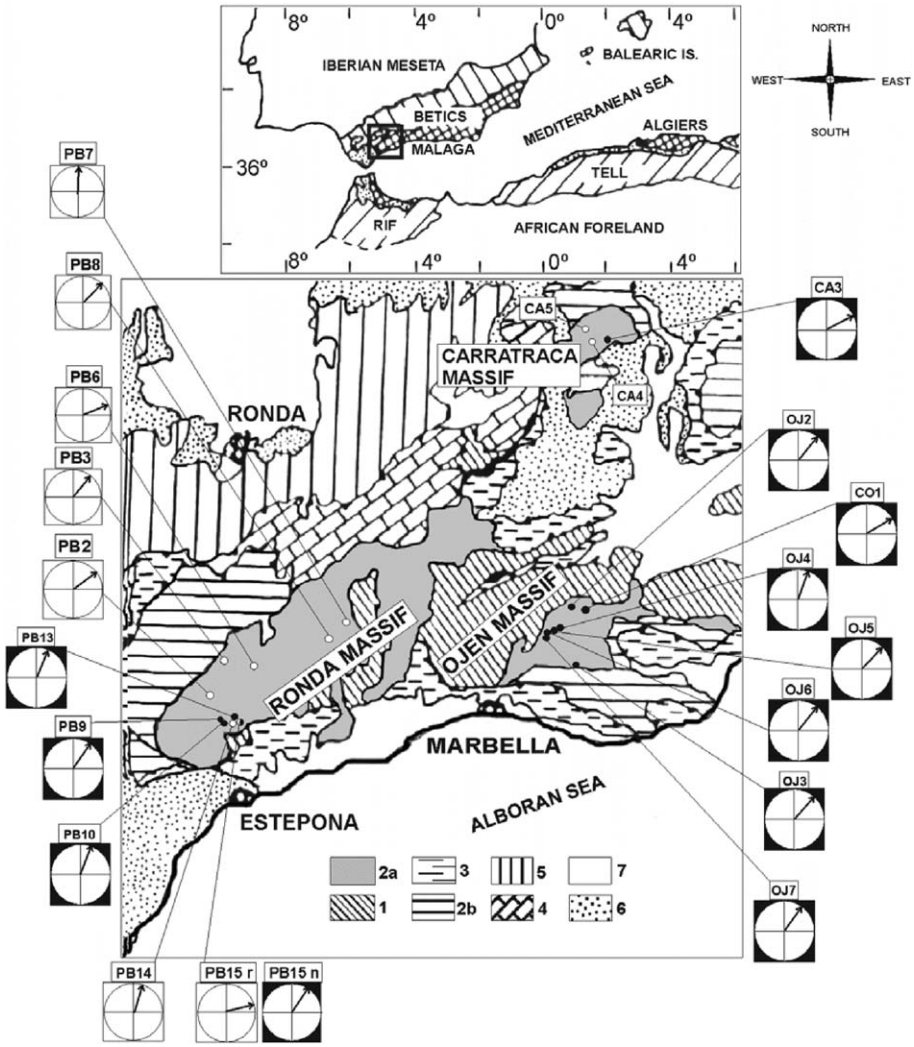


Fig. 1. Simplified tectonic map of the western Alpujárrides showing the location of the sampled sites, the amount of block rotations reflected by the ChRM in each of the studied sites and its polarity (black background = normal polarity; white = reversed polarity). *Western Alpujárriddes:* (1) Blanca unit; (2a) Ultramafic massifs; (2b) Casares unit. *Others:* (3) Maláguide complex; (4) Dorsal complex; (5) SubIberian margin rocks (external zones); (6) Flysch and Alozaina complexes; (7) post-orogenic sedimentary rocks (Neogene).

found. A petrographic zonation is observed within the outcrops, from plagioclase lherzolites in the bottom and spinel lherzolites at intermediate depths to garnet-bearing peridotites in the top. This zonation is also reproduced within the inter-bedded mafic layers (Obata, 1980) that constitute up to 5% in volume of the peridotitic massifs. This zonation has been reinterpreted (Van der Wal and Vissers, 1993) in terms of microstructural data, showing that the massifs consist of a granular domain in the middle of the bodies bordered by a spinel tectonites domain (to the top) and a plagioclase tectonites domain (to the bottom). Toward the contacts with the metamorphic cover the porphyroclastic microstructure of the spinel tectonites evolves to mylonitic.

The earlier uplift and emplacement models proposed for the western Mediterranean peridotites involved diapiric up-rise of mantle rocks (Loomis, 1972, 1975; Obata, 1980). However, Lundeen (1978) demonstrated that the Ronda peridotites form

an allochtonous thrust sheet instead of a mantle diapir rooting in the present-day upper mantle. Geophysical data agree with this conclusion (Torné et al., 1992). Since then, general agreement exists on the solid-state emplacement of the peridotites into the crust.

Regarding the tectonic scenario and the age of emplacement of the peridotites within the mid-crustal rocks, different models have been proposed. Tubía and Cuevas (1986, 1987) and Vauchez and Nicolas (1991) considered the emplacement of the western Mediterranean peridotites in a rift undergoing strike-parallel motion. The entire process is considered to be of Alpine age (Tubía and Cuevas, 1986). Balanyá and García-Dueñas (1987) envisaged the emplacement of the peridotite bodies by large scale thrusting as a result of Paleogene contractional events, preceded by extensional tectonics to explain the lack of lower crust above the peridotites. These events took place before the overthrusting of the Alborán domain over the SudIberian and Maghrebian domains in the Gibraltar Arc. Later, during the early Miocene, the peridotite slab was extensionally dismembered to form individual bodies, some of which now outcrop.

Regardless of their origin, the western Mediterranean peridotites, together with the juxtaposed crustal units, suffered a high-temperature, high- to medium-pressure metamorphic evolution (Loomis, 1975; Seideman, 1976; Priem et al., 1979; Michard et al., 1983; Zindler et al., 1983; Polvé, 1983; Reisberg et al., 1989, 1991; Sánchez-Rodríguez and Gebauer, 2000, amongst others). The detailed study of Sánchez-Rodríguez and Gebauer (2000) established that the temperature peak of the metamorphism was 790 ± 15 °C, took place 19.9 ± 1.7 Ma ago and was followed by a rapid cooling up to 17–18 Ma ago, with a cooling rate between 200 and 340 °C per million years. The post-metamorphic cooling of the peridotites gave rise to the serpentinization of most of the igneous assemblages and the generation of a first population of submicroscopic magnetite (Gervilla, 1990). This alteration process proceeded at the end of the cooling history of the peridotites, between approximately 20 and 17–18 Ma ago. This age will be considered as the age of the acquisition of the characteristic remanent magnetization (ChRM) of the peridotites. Later, during the extensional dismembering of the peridotite slab, a new alteration assemblage made up of chrysotile with millimetre-size magnetite grains replaces the earlier, lizardite-based assemblage, along fractures and joints. This alteration process continued up to the present.

3. Sampling strategy and palaeomagnetic methods

A total amount of 23 ultrabasite sites were sampled, distributed through the Ronda, Ojén and Carratraca massifs (Fig. 1). Different peridotites lithologies were sampled as well as serpentinized peridotites. Field work was organized in two field campaigns. In the first phase, nine sites were sampled: PB2, PB3, PB6, PB7, PB8, CA3, CA4, CA5 and OJ2 (PB = Peñas Blancas, Ronda massif; CA = Carratraca massif; OJ = Ojén massif). The lithology of these nine sites varied between fresh peridotites to highly serpentinized ultrabasites. After a pilot palaeomagnetic study, it was observed that palaeomagnetic directions appeared better determined and isolated in fresh peridotites, whereas in serpentinized peridotites, present-day field directions or unstable behaviour was observed. Therefore, the second phase of the campaign was directed toward the sampling of different lithologies of fresh peridotites (harzburgites, lherzolites, dunites, spinel lherzolites–tectonites, pyroxenites and plagioclase lherzolites). Fourteen sites were sampled, with 11 giving useful palaeomagnetic information: PB9, PB10, PB13, PB14, PB15, OJ3, OJ4, OJ5, OJ6, OJ7 and CO1 (PB = Ronda massif; OJ = Ojén massif; CO = Coín, Ojén massif).

Palaeomagnetic measurements were carried out at the palaeomagnetic laboratory of Madrid and rock magnetic analyses were performed in the laboratories of Madrid, Scripps, Utrecht and Zurich. The NRM was measured with a JR-5A spinner magnetometer (Agico). Progressive thermal and alternating field (AF) demagnetization were performed using a Schonstedt TSD-1 furnace and a Schonstedt GSD-5 AC alternating field demagnetizer, respectively. Bulk magnetic susceptibility was measured after each thermal demagnetization step to monitor magnetic mineral alteration produced by heating. In addition, some rock magnetic parameters were measured: bulk

magnetic susceptibility (Kappabridge KLY-3 Agico); hysteresis cycles (VSM Micromag and Coercivity Spectrometer); Curie curves (Curie Balance); and isothermal remanent magnetization (IRM) acquisition curves (Impulse Magnetizer ASC Scientific IM10-30). Pilot samples were subjected to thermal demagnetization of three orthogonal IRM components (Lowrie, 1990).

4. Rock magnetic results

In addition to the examination of NRM magnetic components, several rock magnetic experiments have been carried out in order to constrain the magnetic mineralogy of the samples.

4.1. Bulk magnetic susceptibility

Bulk magnetic susceptibility has been measured for 240 peridotite samples of different lithologies and for 23 serpentinites. Despite the variability in susceptibility values within each lithology, two distinct populations can be observed in the distribution of susceptibility values (Fig. 2; Table 1). The magnetic susceptibility is greater in serpentinites than in peridotites, indicating a higher proportion of ferromagnetic (sensu lato) minerals in samples affected by the serpentinization. The degree of serpentinization in the Ronda peridotites is usually small (< 30% of the total rock volume), and it is concentrated near the contact and fractures zones, but incipient serpentinization can be found even in the fresh peridotites. Site CA5 (lherzolites with an intermediate degree of serpentinization) deviates from this general pattern. Its low susceptibility is consistent with peridotite behaviour, while its NRM behaviour (described below) is similar to that of the serpentinites. This site has not been included in Fig. 2.

4.2. Curie curves

The variation of saturation magnetization with temperature for different samples shows clearly the predominant role of magnetite in the magnetic mineralogy of the studied sites (Fig. 3). Heating and cooling was carried out in air, and irreversible magnetic changes are evidenced by the different behaviour of heating and cooling curves. Magnetite seems to be destroyed by heating and a new phase with a Curie temperature of 350 °C is created. A small hump in the heating branch around 250 °C is observed in PB12 and OJ7 samples (Fig. 3). If this feature is not an artefact due to instrumental noise, it could be related to the presence of hexagonal pyrrhotite, since this mineral exhibits the so-called λ transition (Dunlop and Özdemir, 1997).

4.3. IRM experiments

IRM acquisition curves were obtained for 12 samples of fresh peridotites (Fig. 4). The samples

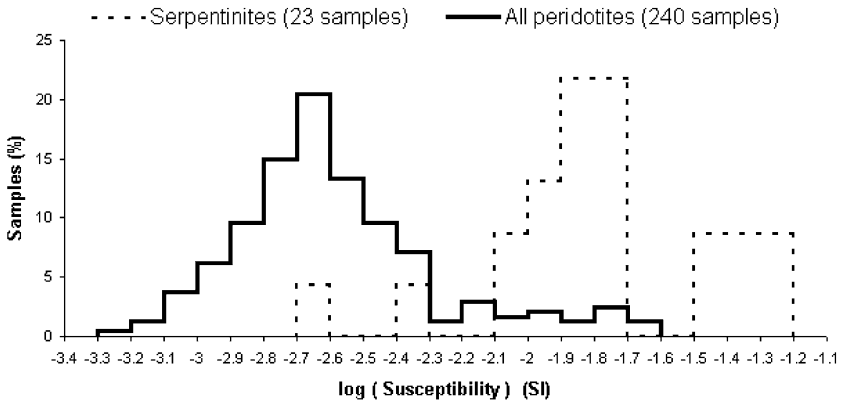


Fig. 2. Bulk magnetic susceptibility distribution. In abscises, the logarithm of the susceptibility (in SI units). In ordinates, the percentage of samples with susceptibilities between two given values.

Table 1
Bulk magnetic susceptibility for different lithologies

Lithology	Sites	Number of samples	Mean susceptibility (10^{-3} SI)	Maximum value (10^{-3} SI)	Minimum value (10^{-3} SI)	Standard deviation (10^{-3} SI)
Spinel lherzolites (tectonites)	CO1, OJ3, PB3	43	2.0	4.4	0.9	1.0
Plagioclase lherzolites	OJ4, OJ5, PB15	49	5.2	24.5	0.8	6.0
Lherzolites	OJ2, PB6, PB8, PB9	53	2.5	6.9	0.6	1.1
Dunites	OJ6, OJ7	34	4.1	8.9	1.8	1.7
Harzburgites	PB2, PB10, PB12, PB13	61	4.1	20.9	1.2	5.1
Pyroxenites	PB11, PB14	25	1.8	7.6	1.0	1.3
Serpentinites	PB7, CA3, CA4	23	21.8	62.3	2.1	16.2
Intermediate serpentinized lherzolites	CA5	10	1.2	3.3	0.4	1.0

Mean, maximum, minimum and standard deviation susceptibility values for each lithology.

reach saturation below 0.3 T, indicating the dominant presence of low coercivity minerals. No indication of high coercivity phases (i.e., haematite) is found in these data. Thermal demagnetization of three orthogonal IRM components indicates that the main carrier of the remanence presents a maximum unblocking temperature of 575–600 °C, pointing to the presence of magnetite. Smaller drops in intensity can be observed at lower temperatures (225–250 °C) in some samples. This could be due to the presence of a different low coercivity magnetic phase or could be related to the grain size distribution of the magnetite.

4.4. Hysteresis cycles and parameters

Hysteresis cycles were obtained for 25 samples of "fresh" peridotites and for 1 serpentinite sample. In Fig. 5, two examples of hysteresis cycles, after correction for paramagnetic contribution, are represented. A Day plot (Day et al., 1977) of the investigated samples is shown in Fig. 6. Considering

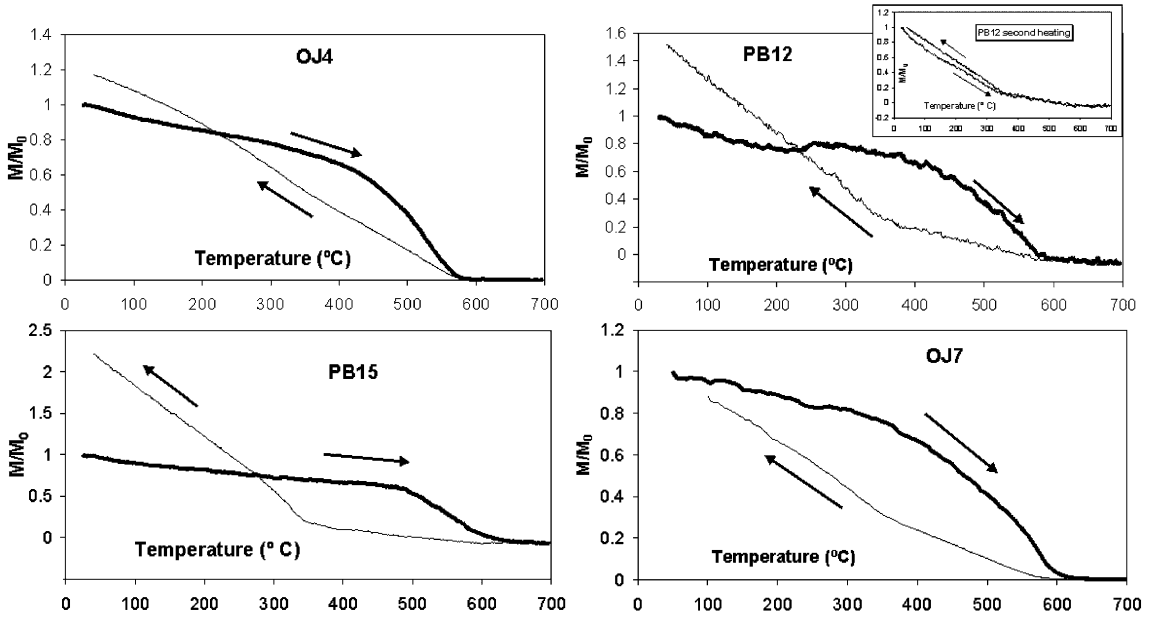


Fig. 3. Thermomagnetic curves of different lithologies of periodotites: OJ4 and PB15, plagioclase periodotites; PB12, harzburgite; OJ7, dunite.

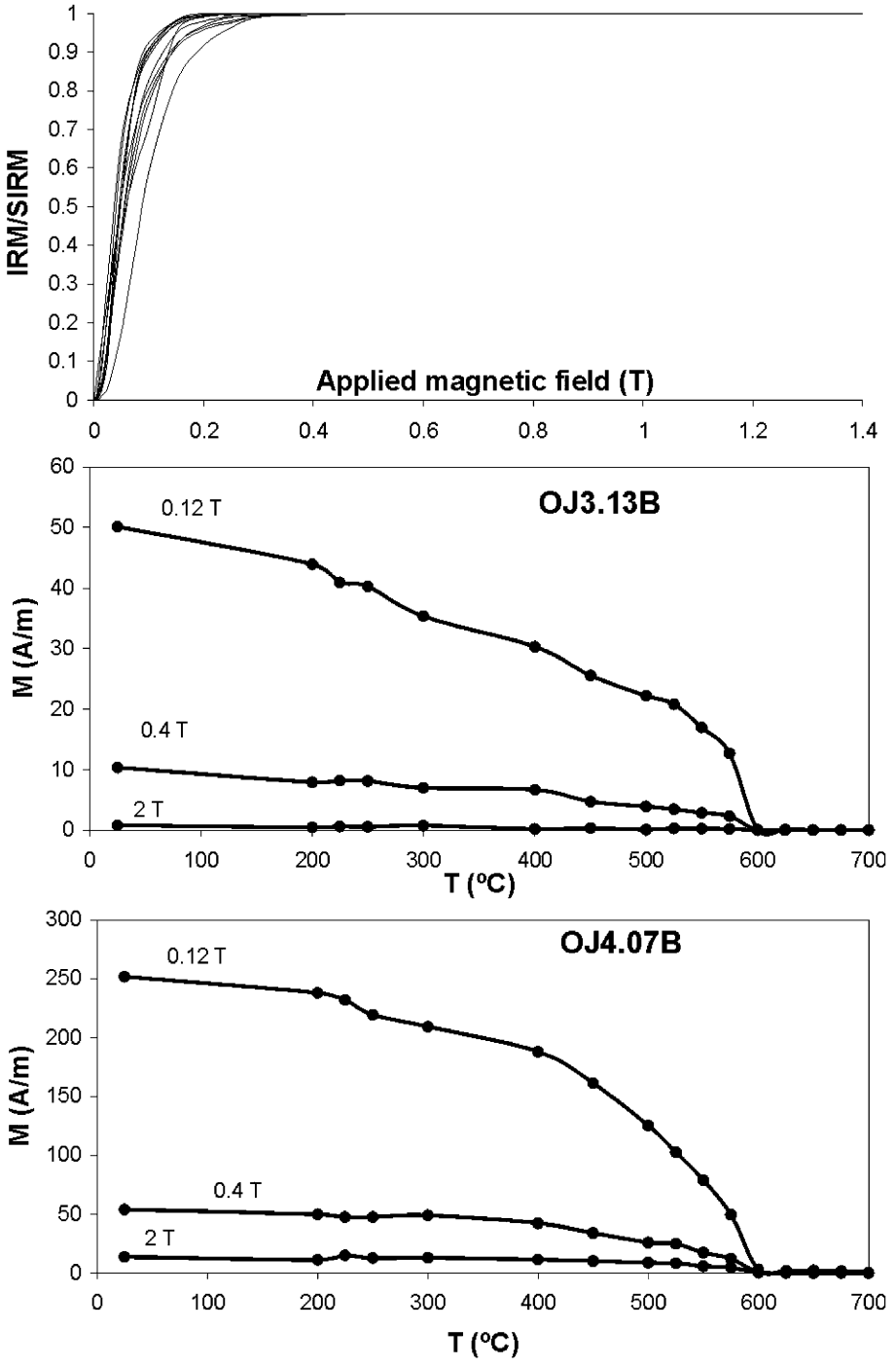


Fig. 4. Normalized IRM acquisition curves for periodotites (top); thermal demagnetization curves of the three orthogonal IRM components for two representative periodotite samples.

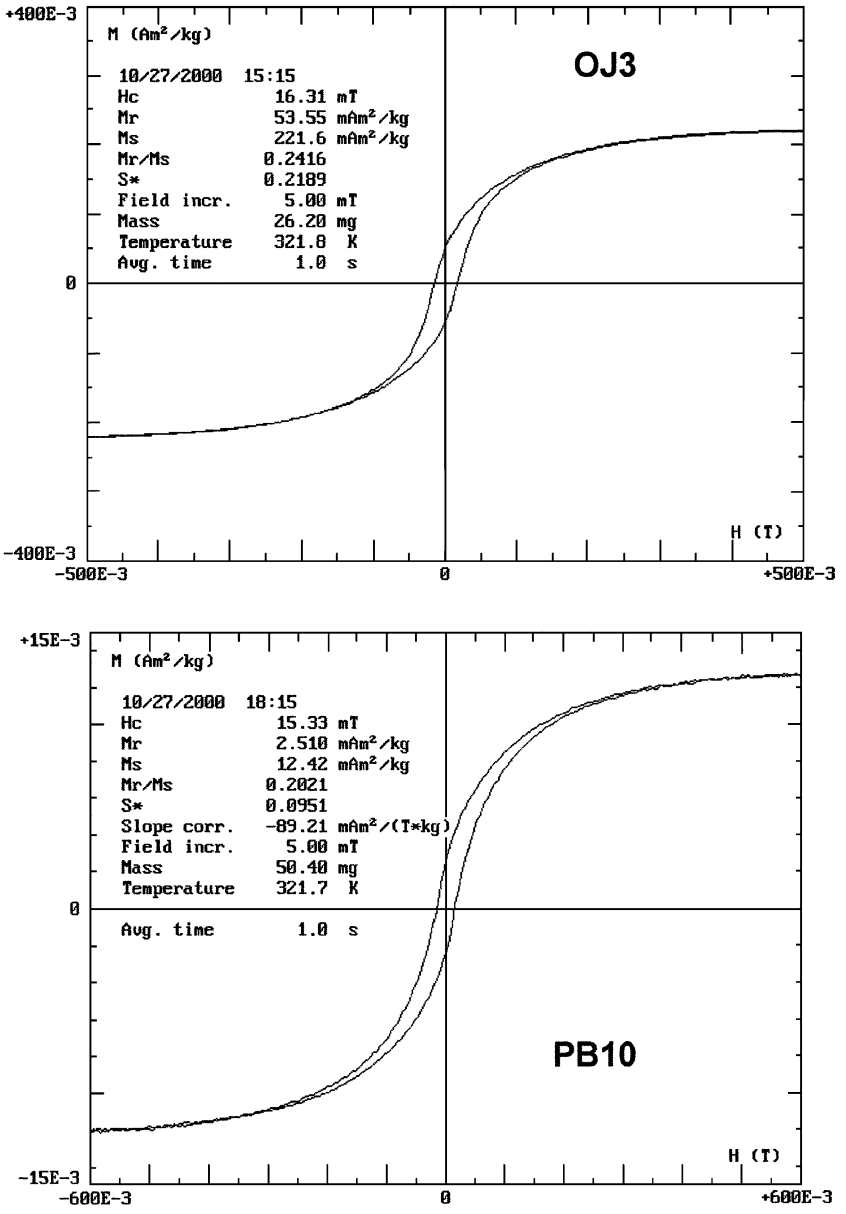


Fig. 5. Hysteresis cycles for OJ3 (spinel 1herzolite, *top*) and PB10 (harzburgite, *bottom*).

the recent works of Dunlop (2002a,b), the data from fresh peridotites fall in the region where theoretical calculations and empirical data situate the PSD/SD + MD (pseudo-single domain/single domain + multi-domain) trends. The data show more dispersion, indicating a broader distribution of sizes and/or compositions. Specifically, pyroxenite samples seem to follow a different pattern and are located far away from the other samples. Our results in "fresh" peridotites are consistent with the data obtained in intrusive rocks from the oceanic crust (shown in Dunlop, 2002b). In addition, the serpentinized sam-

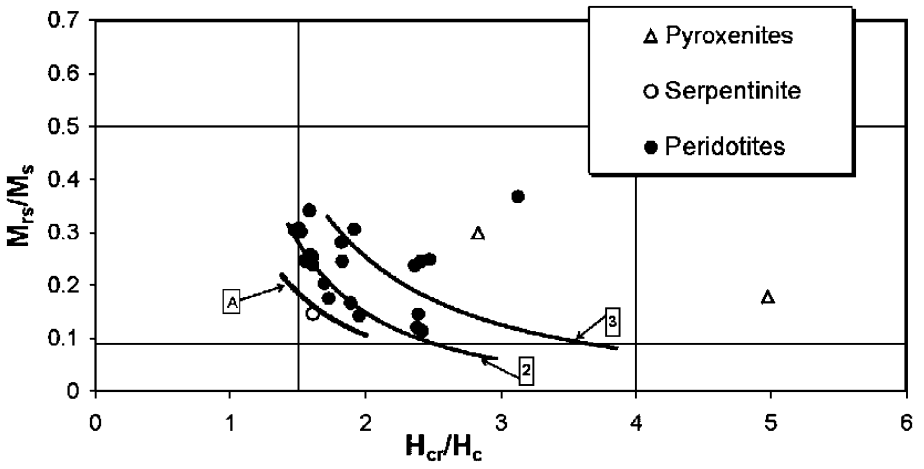


Fig. 6. Hysteresis parameters of investigated samples. Data from Dunlop (2002b) are also showed. (A) Experimental data compiled by Dunlop (2002b) from serpentinized periodotites. (2 and 3) Theoretical curves for SD and MD elongated magnetite mixtures (Dunlop, 2002b).

ple falls in the empirical curve estimated by Dunlop (2002b) from serpentinized peridotites.

In conclusion, rock magnetic experiments indicate the presence of a mixture of SD and MD magnetite, sufficient to carry a stable palaeomagnetic signal over geological times. The presence of a mineralogical phase with a characteristic temperature of 250 °C cannot be excluded, but can neither be clearly demonstrated.

5. Palaeomagnetic results

Detailed thermal and alternating field demagnetization experiments were carried out on pilot samples. Both techniques were effective in isolating stable components, but more linear demagnetization paths were obtained by thermal treatment. Therefore, most samples were systematically demagnetized by heating from room temperature up to 200 °C and then up to

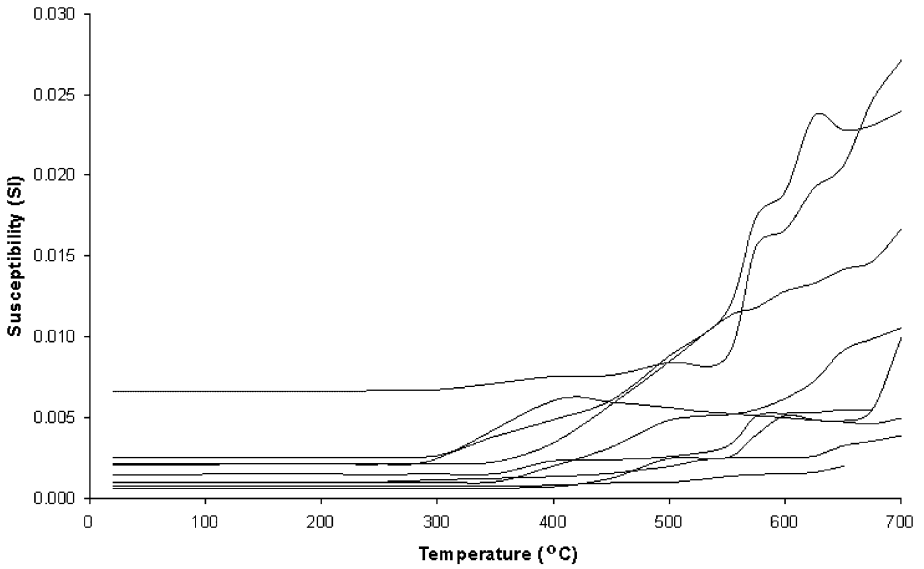


Fig. 7. Variation of bulk magnetic susceptibility during thermal demagnetization.

600 °C, in temperature increments ranging from 25 to 100 °C. After each step, bulk magnetic susceptibility was measured in order to identify thermally induced magnetic alteration. These measurements show that most samples experienced chemical changes above 400 °C. These changes are responsible for the progressive increase in bulk magnetic susceptibility, observed in Fig. 7. Generally, in spite of these chemical changes, the remanent magnetization exhibited a linear demagnetization path. When the alteration produced by heating started at lower temperatures, this alteration affected the magnetization direction and the results of AF demagnetization were preferred.

The samples can be divided into four groups according to their magnetic behaviour during NRM demagnetization (see Table 2). Group 1: samples that after cleaning a viscous component ($T_b \leq 200$ °C) exhibited only one stable magnetic component with a direction different from the present-day field. This component was considered as the ChRM. Group 2: samples with two magnetic components, a first northward-directed component (with a T_b>200 °C) and the ChRM. Group 3: samples showing only a present-day field direction. Group 4: samples characterized by an unstable behaviour during demagnetization or anomalous directions.

5.1. Group 1

Most samples from the ''fresh'' peridotite sites belong to this first group, including 75%–100% of samples from sites PB2, PB13, PB14, CO1, OJ2, OJ3 and OJ4; and 40%–75% of samples from PB9, PB10, PB15, OJ5 and OJ7. Mean initial NRM values range from 0.01 up to 0.15 A/m, with the exception of PB10 and OJ4, which present higher values between 0.73 and 1.08 A/m. A stable directional component is

Table 2
Number of samples belonging to the different palaeomagnetic behaviour groups for each investigated site, with mean values of initial NRM intensity and bulk magnetic susceptibility

Site	Lithology	N	Group 1			Group 2			Group 3			Group 4		
			N1	NRM (A/m)	χ (10^{-3} SI)	N2	NRM (A/m)	χ (10^{-3} SI)	N3	NRM (A/m)	χ (10^{-3} SI)	N4	NRM (A/m)	χ (10^{-3} SI)
PB2	H	15	14	0.15	2.1	0	–	–	0	–	–	1	2.45	2.3
PB3	SL	16	7	0.02	2.3	1	0.03	1.3	7	0.60	2.9	1	0.05	3.0
PB6	L	19	5	0.016	2.3	1	0.02	2.7	10	0.20	2.0	3	0.10	5.5
PB7	SP	8	0	–	–	7	1.00	41.0	1	1.10	41.6	0	–	–
PB8	L	18	0	–	–	7	0.05	3.0	11	0.10	3.0	0	–	–
PB9	L	10	5	0.010	2.1	2	0.03	2.5	2	0.15	3.7	1	0.02	3.7
PB10	H	12	5	1.08	14.8	0	–	–	0	–	–	7	0.07	2.3
PB13	H	12	11	0.04	2.2	0	–	–	0	–	–	1	0.009	1.2
PB14	PX	10	10	0.04	1.4	0	–	–	0	–	–	0	–	–
PB15	PL	12	8	0.03	2.7	3	0.015	1.4	0	–	–	1	0.005	1.9
CO1	SL	12	10	0.01	1.3	0	–	–	0	–	–	2	0.004	1.2
OJ2	H	13	9	0.07	1.2	3	0.12	0.9	1	0.58	3.6	0	–	–
OJ3	SL	12	9	0.10	2.2	0	–	–	0	–	–	3	0.10	3.3
OJ4	PL	12	11	0.73	11.9	0	–	–	0	–	–	1	0.09	11.4
OJ5	PL	12	5	0.05	1.3	0	–	–	0	–	–	7	0.003	0.9
OJ6	D	12	1	0.03	3.5	6	0.04	4.6	0	–	–	5	0.03	3.3
OJ7	D	12	7	0.05	4.5	4	0.05	5.4	0	–	–	1	0.02	2.4
CA3	SP	10	0	–	–	6	0.55	12.9	2	0.70	11.6	2	1.01	16.2
CA4	SP	11	0	–	–	1	0.23	17.3	10	0.80	12.6	0	–	–
CA5	L + SP	15	0	–	–	5	0.03	0.6	9	0.17	1.6	1	0.01	–
Total N's		253	117			46			53			37		

The groups are as follows: Group 1, samples with a sole rotated component; Group 2, samples with a first component with a northward direction and a second rotated component; Group 3, samples with a sole northward component; and Group 4, samples with nonsystematic or unstable behaviour. Lithology: L = lherzolites; H = harzburgites; D = dunites; PL = plagioclase lherzolites; SL = spinel lherzolites; PX = pyroxenites; SP = serpentinites; L + SP = serpentinized lherzolites.

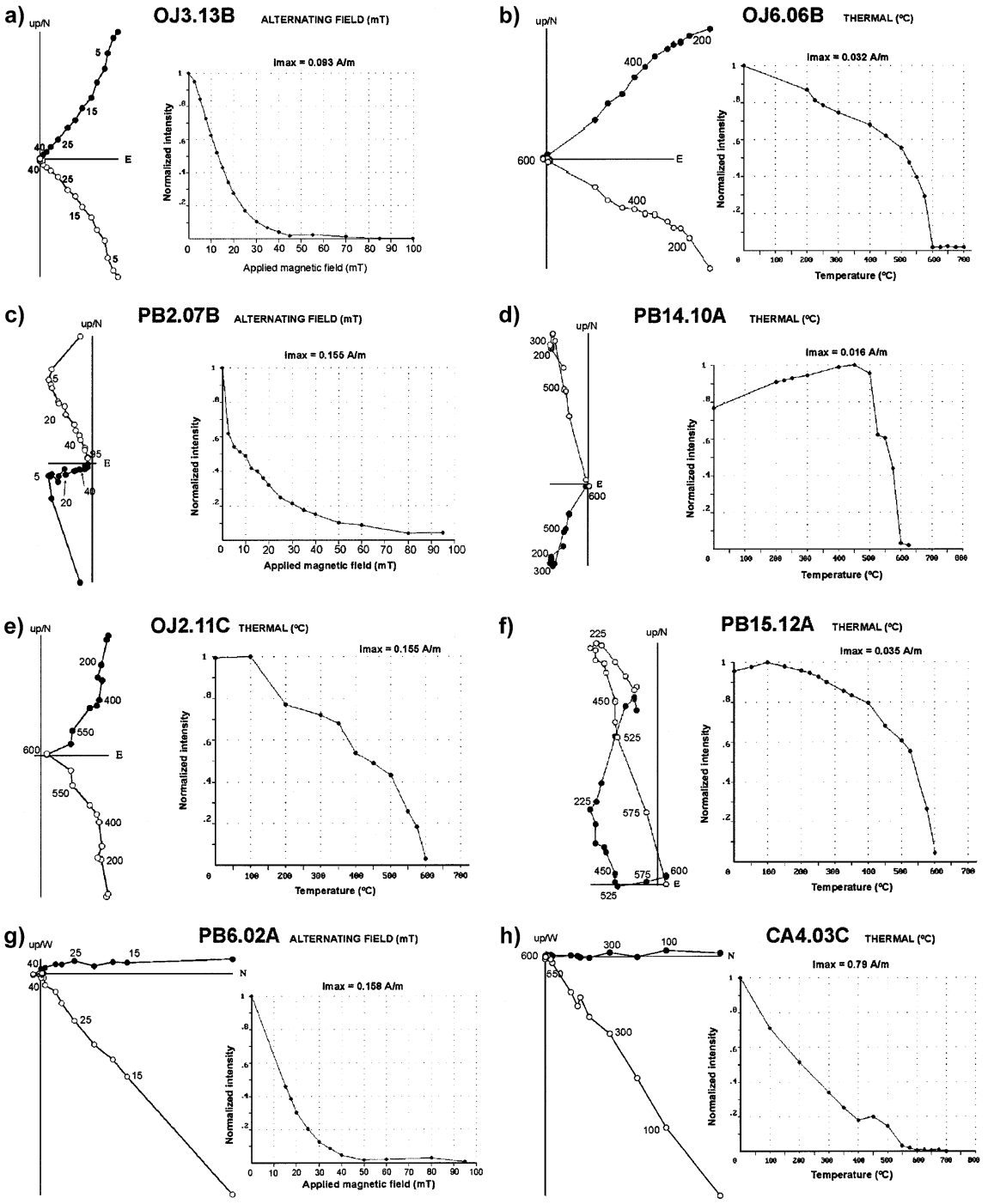


Fig. 8. Zijderveld and intensity decay plots for representative samples of different groups: (a) alternating field (AF) demagnetization, group 1, normal polarity; (b) thermal (TH) demagnetization, group 1, normal polarity; (c) AF demagnetization, group 1, reversed polarity; (d) TH demagnetization, group 1, reversed polarity; (e) TH demagnetization, group 2, normal polarity; (f) TH demagnetization, group 2, reversed polarity; (g) AF demagnetization, group 3; (h) TH demagnetization, group 3.

easily isolated during both AF and thermal demagnetization. A magnetic phase of low coercivity (median destructive fields between 5 and 25 mT) and maximum unblocking temperatures of 550–575 °C (Fig. 8a, b, c and d) suggests the presence of magnetite as the carrier of the magnetization, in agreement with the results of rock magnetic experiments. The ChRM gives normal and reversed polarities, depending on the sites. Generally, samples present a common polarity within a site, with the exception of sites PB15 and CA3, where both polarities have been obtained. At PB15, contrasting polarities were observed in two groups of specimens, sampled in the same outcrop but separated by a few metres. This site was divided into two sites depending on the polarity of the ChRM (PB15n and PB15r). At CA3, one sample presented reversed polarity and four normal polarity. Considering the compositional layering at these two outcrops, the polarity of samples seems to be consistent with its "layer" position. That is to say, samples from the same "layer" present the same polarity. Where two polarities of the ChRM are observed in one site (PB15 and CA3), samples with the same polarity are grouped geographically (at the outcrop scale) and separated by centimetres or few metres from samples that exhibited the other polarity. The ChRM directions were well grouped (in situ) and presented NE or SW directions (Table 3).

5.2. Group 2

Group 2 comprises about 20% of the total amount of samples, mostly from sites PB7, PB8, OJ6 and CA3, which exhibited moderate to high degree of serpentinization. The initial NRM intensity ranged between 0.02 and 0.55 A/m. Stepwise demagnetization shows the presence of two magnetic components. A low-temperature component with a present-day field direction and unblocking temperatures of about 350–450 °C, and a high-temperature component with

Table 3
Calculated directions and statistical confidence parameters for ChRM and for the mean of all directions

Site	Lithology	N	Before "tectonic" correction				After "tectonic" correction			
			D (°)	I (°)	α_{95}	k	D (°)	I (°)	α_{95}	k
ChRM										
PB2	H	9	238.4	−62.0	7.7	45.1	268.1	−33.9	7.7	45.1
PB3	SL	7	224.4	−38.0	12.3	25.2	182.1	−16.6	12.9	22.8
PB6	L	4	252.9	−41.4	21.5	19.2	168.9	−16.7	51.3	4.2
PB7	SP	7	187.3	−31.9	15.9	15.4	191.3	−56.6	15.9	15.4
PB8	L	8	228.7	−56.5	21.7	7.5	257.5	−71.7	21.7	7.5
PB9	L	8	39.7	36.8	6.1	82.4	20.6	6.1	6.1	82.4
PB10	H	5	26.0	41.3	8.8	76.2	72.3	19.8	8.8	76.2
PB13	H	4	27.5	37.6	20.2	21.7	16.4	−10.4	20.2	21.7
PB14	PX	10	201.7	−64.0	4.3	129.2	173.4	−3.3	4.3	129.2
PB15normal	PL	5	38.8	45.1	11.7	43.6	26.1	−2.8	11.7	43.6
PB15reversed	PL	5	261.8	−65.6	15.5	25.3	201.8	−27.8	15.5	25.3
CO1	SL	10	64.3	42.4	7.8	39.5	55.5	3.8	7.8	39.5
OJ2	H	7	43.8	42.3	8.6	49.9	9.1	30.7	8.6	49.9
OJ3	SL	10	46.3	46.0	6.9	49.5	111.5	75.9	6.9	49.5
OJ4	PL	11	24.9	44.8	3.7	150.7	108.8	62.5	3.7	150.7
OJ5	PL	5	48.0	30.9	14.7	28.1	59.0	12.8	14.7	28.1
OJ6	D	6	43.4	26.9	7.4	82.9	71.6	37.7	7.4	82.9
OJ7	D	6	40.7	33.7	8.0	71.1	317.6	72.4	8.0	71.1
CA3	SP	5	67.7	69.9	16.3	22.9	56.8	28.1	16.1	23.6
CA4	SP	0				–				
CA5	L + SP	0				–				
Mean		19	45.3	45.8	7.6	20.4	34.9	33.6	18.6	4.2

The "tectonic" correction is based upon the direction of the compositional layering of the peridotites, except in PB6, where a metric-scale fold was sampled and a proper fold test has been applied with negative results. Lithology: L = lherzolites; H = harzburgites; D = dunites; PL = plagioclase lherzolites; SL = spinel lherzolites; PX = pyroxenites; SP = serpentinites; L + SP = serpentinized lherzolites.

similar characteristics to the ChRM observed in group 1: low coercivity, maximum unblocking temperatures of 550–575 °C, samples with normal and reversed magnetizations and NE and SW directions (Fig. 8e,f). A few samples from this group were not completely demagnetized at 600 °C, indicating a small contribution of haematite. It was not possible to detect any systematic behaviour of the small component associated to this mineral, and, therefore, this component has not been considered in the discussion.

5.3. Group 3

Group 3 samples exhibit a single component with a direction parallel to the actual geomagnetic field (Fig. 8g,h). Initial NRM intensity varied from 0.1 up to 0.8 A/m. Most samples belonging to this group are found in sites that have a high degree of serpentinization, such as site CA4 where 90% of investigated samples exhibited this behaviour. Within a site, it was easy to detect the samples of this group by their high values of initial NRM intensity. For example, in site PB3, where samples belonging to group 1 and group 3 were found, samples that presented initial NRM intensities of around 0.60 A/m exhibited a northward direction (group 3), while samples with initial NRM intensity values of 0.02 A/m behave as group 1.

5.4. Group 4

The 15% of samples, falling within this group, were rejected because scattered directions were found within site or because an anomalous behaviour was observed during demagnetization.

6. Discussion

6.1. Origin of the ferromagnetic minerals

It is widely accepted that in peridotitic bodies, the generation of the ferromagnetic minerals (usually magnetite) is associated to serpentinization processes (Dunlop and Prévot, 1982; Bina and Henry, 1990). In the case of the Ronda peridotites, the serpentinization history can be divided in two main phases (Gervilla, 1990). The first one took place during the post-metamorphic cooling of the massifs. This cooling was simultaneous to the decompression of the peridotites, which, in turn, was connected to an increase in porosity and permeability and hence to an increase in fluids circulation. The serpentinization process could start below 550 °C, the thermal stability limit of most serpentine minerals, but in the Ronda peridotites, the predominance of lizardite (locally with brucite) over the serpentine polymorphs indicates that serpentinization started at lower temperatures (tentatively around 350–400 °C). Under these conditions, not all the iron from the primary olivine is used to form magnetite but part of it enters the structure of lizardite and brucite (Moody, 1976). Microscopic studies carried out in the Ronda peridotites indicate that in samples with up to 30% in volume of serpentine, magnetite has a sub-microscopic grain size (<50 μm), and it is not very abundant (Gervilla, 1990). This sub-microscopic magnetite constitutes a first population of ferromagnetic minerals contributing to the NRM. Taking into account the thermal history of the Ronda peridotites, we have considered that the age of the genesis of this magnetite is between 20 and 17–18 Ma.

A second serpentinization phase was associated to the extensional dismembering of the peridotite slab. This alteration proceeded mainly along the fractures and joints developed during this extensional phase, and continued up to the present days. During this process, the mineral assemblages altered to chrysotile (low-temperature and low-pressure polymorph of serpentine). Since iron is not easily incorporated into the chrysotile structure, the iron from the olivine, and from the lizardite and the brucite formed during the previous phase, was concentrated in the form of a coarse-grain population of magnetite that developed between the chrysotile fibres with sizes up to the millimetre range (Gervilla, 1990). The main characteristics of this second population of magnetite are its abundance and coarse grain size.

Some thermomagnetic curves suggest the possible presence of pyrrhotite. This phase is considered a high-temperature igneous phase present in the peridotites. Nevertheless, its importance as a magnetic carrier has been shown to be less than marginal, since in the studied outcrops, the magnetic remanence is undoubtedly carried by magnetite. In some samples, there are evidences of the presence of haematite. Serpentinization is a highly reducing process, therefore, any haematite had to be formed in younger

times. Again, its palaeomagnetic importance is found to be very low in the studied sites.

In summary, from the point of view of the serpentinization history of the peridotites, two distinct populations of magnetite grains are expected to carry a magnetic remanence: (1) a primary population of fine grain size, developed during the post-metamorphic cooling of the massifs; and (2) a secondary population, generated since the extensional dismembering of the massifs to the present days, with a coarse grain size and high abundance. This is in agreement with the single-domain and multi-domain (SD and MD) magnetite mixture detected when studying the hysteresis parameters.

6.2. Origin of the magnetic components

Fig. 9 shows the percentage of samples whose northward magnetic component has a relative intensity that falls between two given values, for both peridotites and serpentinites. It can be observed that the intensity of the NRM for the more serpentinized rocks is dominated by the northward component. In constrast, "fresh" peridotites present a very small amount of northward directions. Consequently, the northward-directed component is interpreted as associated to the secondary magnetite population generated during the last serpentinization process. Due to its large grain size, this magnetite behaves as a magnetically "soft" mineral, with its magnetization easily redirected along the present Earth's magnetic field. Therefore, the northward-directed component is interpreted as a viscous magnetization associated to a magnetite population with a multi-domain magnetic behaviour. The variable blocking temperatures of this component can be explained by differences in the relative contribution of this secondary multi-domain magnetite, with respect to the primary population of fine grain magnetite. Moreover, the high abundance of this secondary magnetite in the serpentinized peridotites explains the higher intensities observed in samples that present a northward component.

Samples of groups 1 and 2 were considered useful for palaeomagnetic purposes. The ChRM is associated with the presence of magnetite of high unblocking temperature and stable magnetic behaviour (single- or pseudo-single-domain state). Considering the geological history of the peridotites, the ChRM can be associated mainly to the creation of a sub-microscopic population of magnetite grains during the first serpentinization phase, developed during the post-metamorphic cooling of the peridotites from 350–400 °C to ambient temperatures, between 20 and 17–18 million years ago (Platt et al., 1998; Sánchez-Rodríguez and Gebauer, 2000). Consequently, the ChRM is interpreted as a thermo-chemical remanent magnetization

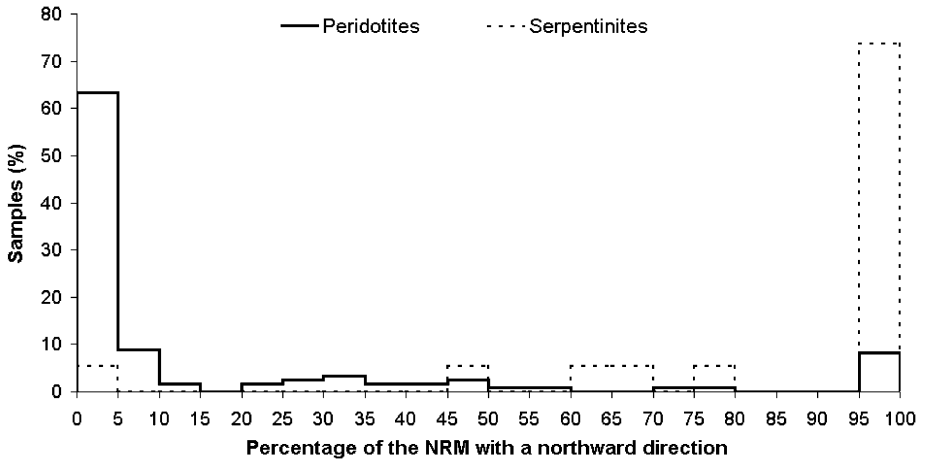


Fig. 9. Percentage of samples whose northward magnetic component has a relative intensity comprised between two given values. More than 70% of the serpentinites have a northward component whose intensity reach 95–100% of their initial NRM intensity, whereas more than 60% of the periodotites have a northward component whose intensity oscillates between 0% (no northward component at all) and 5% of the initial value of NRM.

acquired during the time interval comprised between 20 and 17–18 Ma. The presence of normal and reversed directions indicates that at least one reversal of the Earth's magnetic field was recorded by the peridotites during their cooling.

This interpretation, based on the correlation observed between serpentinization degree and the relative intensity of the northward magnetic component, is in contrast with that of Feinberg et al. (1996). These authors considered the northward component as a partial thermoremanent magnetization (pTRM) acquired during the last steps of the post-metamorphic cooling of the peridotites. Differences in the blocking temperature spectrum of samples belonging to group 1 (Fig. 8b,d) and group 3 (Fig. 8h) seem to favour our magnetic interpretation.

Finally, the presence of a relative high proportion of the northward component in some apparently fresh peridotites could indicate the existence of an incipient second serpentinization phase.

6.3. Tectonic rotations

Palaeomagnetic results obtained from the analysis of ChRM directions are compiled in Table 3 and plotted in Fig. 10. A good grouping of in situ directions can be observed. Mean site directions have been compared with the expected Oligocene–Miocene boundary palaeodirection for stable Iberia (D=4.1°; I=51.4°; α_{95}=5.9; k=246.9; Barberá et al., 1996). Declinations show values that are clockwise rotated by $41 \pm 12°$ if compared with the expected palaeodeclination. In contrast, the inclination values are consistent. This suggests that no important tilting has affected this region since the cooling of the peridotites.

Table 4
Calculated block rotations for each site and the mean value

Site	R (°)	ΔR (°)
PB2	54	16
PB3	40	15
PB6	69	24
PB7	3	17
PB8	45	34
PB9	36	10
PB10	22	12
PB13	23	22
PB14	18	12
PB15n	35	15
PB15r	78	32
CO1	60	12
OJ2	40	12
OJ3	42	12
OJ4	21	9
OJ5	44	16
OJ6	39	11
OJ7	37	11
CA3	64	43
Mean	41	12

The calculation is made comparing the declinations with the expected declination for the Oligocene–Miocene boundary (Barberá et al., 1996).

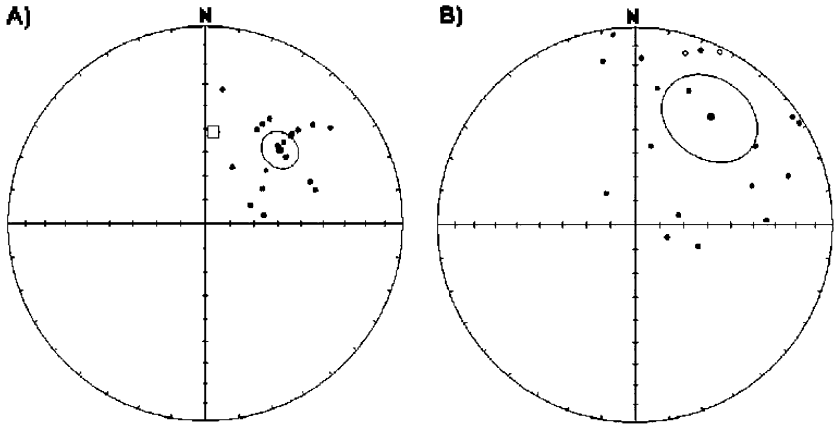


Fig. 10. Stereographic (equal area) projection of ChRM directions. The directions are plotted before (A) and after (B) a fold test based on compositional layering direction. The expected direction for stable Iberia during Oligocene–Miocene boundary (Barberá et al., 1996) is plotted as a white square. The mean and the confidence angle α_{95} are indicated.

The values of the rotations that we have obtained (Table 4; Fig. 2) are in agreement with the palaeomagnetic data obtained by Feinberg et al. (1996) from 10 sites investigated in the Ronda and Ojén Massifs. A similar amount of rotation has been observed in the adjoining western Subbetic zone by Villalaín et al. (1994), and higher rotations in late Oligocene–early Miocene dykes intruding in the Maláguide complex (Platzman et al., 2000; Calvo et al., 2001). Clockwise rotations have been recently observed in the central Subbetic zone that seem to have occurred prior to the rotations observed in the western Subbetic zone (Osete et al., 2004). In order to have a picture of the rotations that have occurred in Tertiary times in central and western Betics, available palaeomagnetic data have been plotted in Fig. 11 and summarized in Table 5. When considering the palaeomagnetic data from the External Betics, rotations detected in both the Jurassic primary component and the Neogene remagnetization have been considered.

In summary, palaeomagnetic data indicate that this region has experienced a clockwise rotation of 41° ± 12° around a vertical axis. The sense of this rotation is in accordance with the results obtained in the central and western Betics.

6.4. Polarity zoning

Palaeomagnetic data demonstrate that the Ronda peridotites recorded a reversal of the Earth's magnetic

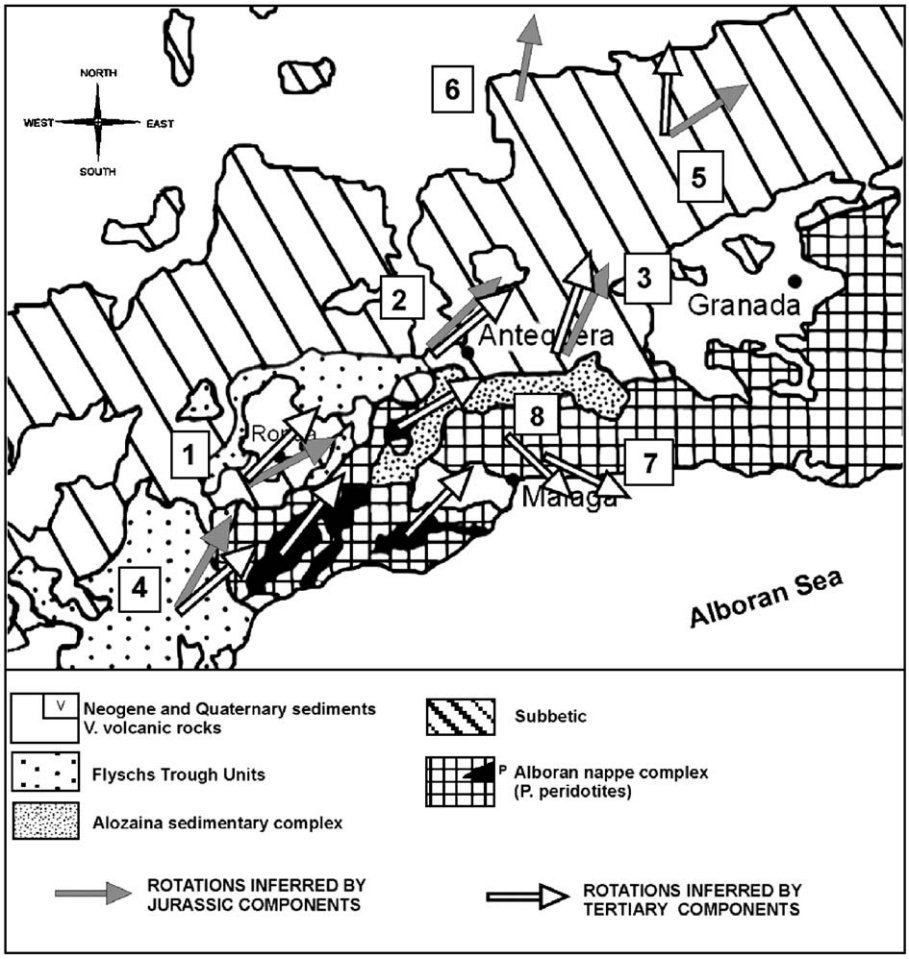


Fig. 11. Block rotations in the Ronda–Málaga region. Data from previous studies have also been included.

Table 5
Previously reported block rotations in the Betic Cordillera

	Localities	N (sites)	Post-Jurassic rotation (°)	Post-Tertiary rotation (°)	Reference
1	JUB	4	62 ± 20	45 ± 37	Villalaín et al. (1994)
	JAT	2			
	JBU	2			
2	JTE	2	46 ± 15	49.5 ± 9.2	Villalaín et al. (1994)
	JTO	4			
3	JGO	2	25.1 ± 5.6	19.1 ± 9.4	Villalaín et al. (1994)
4	JCA	2	30.0 ± 6.8	49 ± 13	Villalaín et al. (1994)
5	CYB	1	57 ± 16	3 ± 15	Osete et al. (2004)
	CYK	1			
	ALJ	2			
	BRJu	1			
	BRJm	1			
	PNS	2			
6	ERJ	2	10 ± 10	–	Osete et al. (2004)
7	MA	8	–	116 ± 18	Calvo et al. (2001)
8	M	8	–	134 ± 10	Platzman et al. (2000)

field during their progressive cooling. This is consistent with the considered age of the ChRM acquisition (between 20 and 17–18 Ma) and the Geomagnetic Polarity time scale (Cande and Kent, 1995). In this time interval, several magnetozones of normal and reversed polarity have been documented (C6n, C5Er, C5En, C5Dr, 5Dn or 5Cr).

Fig. 12 shows the polarity of the ChRM obtained in each of the investigated sites. This figure also includes those data obtained by Feinberg et al. (1996) in which the ChRM is carried by magnetite. In order to homogenize the data, sites from Feinberg et al. (1996) where the polarity was associated with the haematite contribution were discarded. In addition, we have shown the polarity of a few samples from the Carratraca massif where the polarity of the ChRM was well established, but the number of samples was insufficient to statistically calculate a mean direction. These data are not included in Table 3. No relation has been observed between the polarity and the lithology. In the two outcrops, where normal and reversed polarities were observed (PB15 and CA3), samples with different polarities were separated by 1 or 2 m and corresponded to different compositional layers. Therefore, in these places, the magnetic polarity seems to be consistent with a cooling front parallel to the compositional layering.

Normal polarities were systematically found in the Ojén complex. In contrast, normal and reversed polarities were observed in Sierra Bermeja and Carratraca massifs. Therefore, the cooling of the peridotites outcropping in Ojén massif was probably more rapid than those from the Sierra Bermeja or Carratraca outcrops. In addition, if the outcrops studied here are representative of the whole Ojén massif, then palaeomagnetic data suggest that in the Ojén massif the cooling from the upper temperature limit of ChRM acquisition (about 350–400 °C) down to near room temperature took place during no more than 1 Ma, the maximum duration of a normal magnetozone (C6n) in this time interval. This is consistent with the high cooling rates estimated in different regions of the Internal Zones by Sánchez-Rodríguez and Gebauer (2000) and Platt et al. (1998) from radiometric and fission track methods.

As the magnetic polarity can be considered as an isochron, the map of Fig. 12 could be envisaged as a map showing the geochrons of the cooling of the peridotites. A single explanation of the polarity distribution in the Sierra Bermeja and Carratraca massifs cannot be proposed at present with the available palaeomagnetic, petrological and structural data. However, the apparent geographical consistency of the polarity pattern suggests that palaeomagnetism can be a useful tool in understanding the late cooling process of the peridotites.

6.5. Tilt of the compositional layering

In each of the studied sites, the dip of the compositional banding and pyroxenite layers has been measured and a kind of "compositional layering fold-test" has been carried out. Results of this fold test are showed in Fig. 10 and Table 3. The negative result of this test indicates that the compositional layering cannot be considered as a palaeohorizontal at the time of acquisition of the characteristic remanence (i.e., at the cooling time). If this compositional banding was a palaeohorizontal at any older time, then peridotites were deformed and folded at temperatures over 350–400 °C (the upper temperature limit of ChRM acqui-

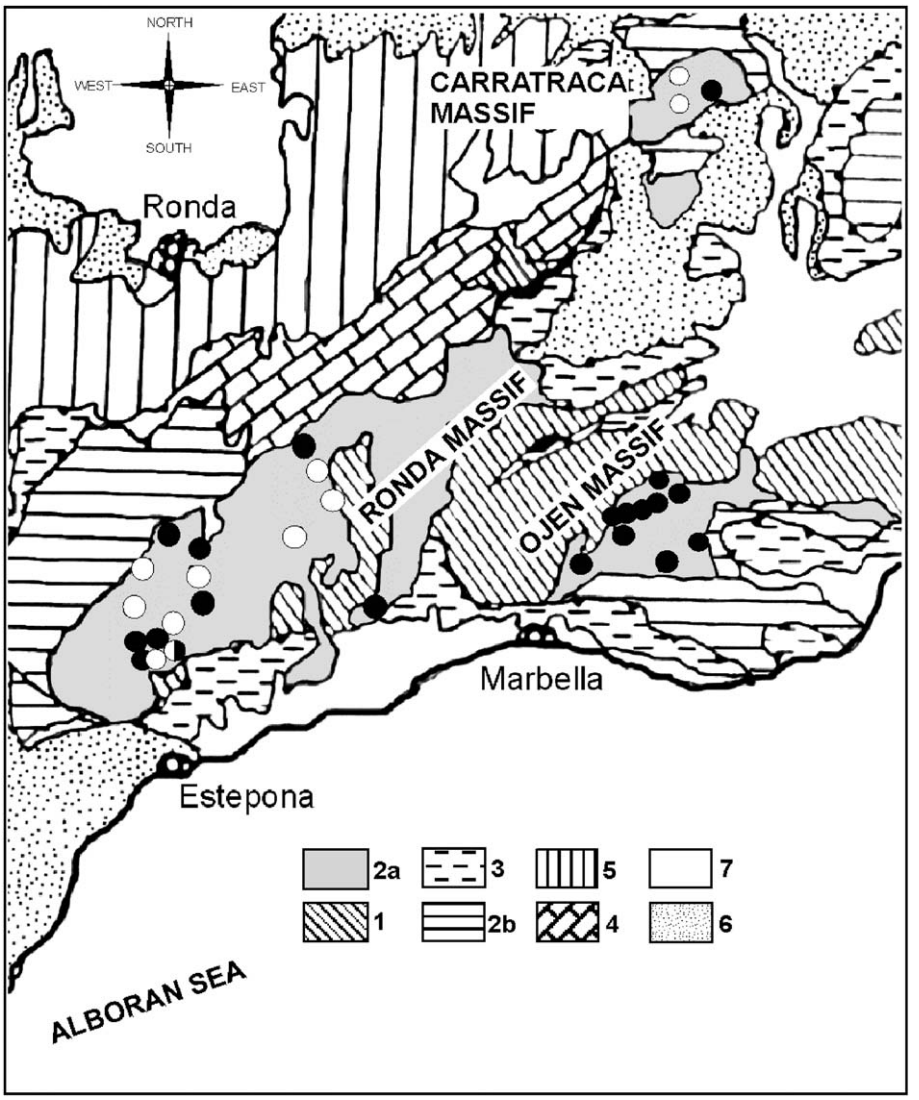


Fig. 12. Polarity of the ChRM of different sites: black circles = normal polarity; white circles = reversed polarity. Data from the study of Feinberg et al. (1996) are also represented. The geological legend is the same as in Fig. 1.

sition). Subsequently, they were cooled, and, later, they experienced clockwise rotations.

6.6. Timing of the rotations

Regarding the amount of the rotations, our results are in agreement with the data of Feinberg et al. (1996), but we disagree on the inferred time of tectonic rotations. Feinberg et al. (1996) suggested that the rotation was completed before the final cooling of the massifs, since some of the samples yielded a magnetic component, stable up to 250 °C, close to the present-day field direction. They considered that this conclusion is also supported by the palaeomagnetic results obtained by Saddiqi et al. (1995) from the Beni–Bousera Massif, where a recent field direction was stable up to 450 °C. In our opinion, from the present palaeomagnetic data is not possible to conclude that rotations were finished prior to the final cooling of the peridotites. In the reasoning of Feinberg

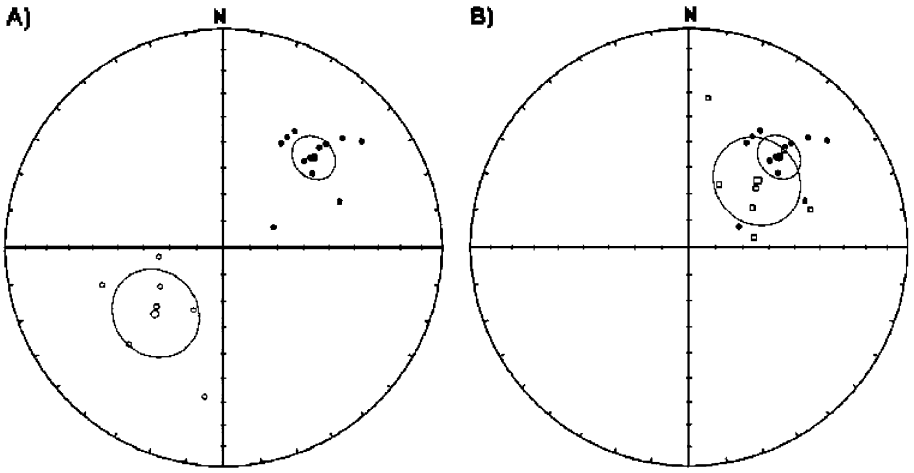


Fig. 13. Normal (black dots) and reversed (white dots) directions of ChRM. Reversal test: (A) in situ directions; and (B) after calculation of antipodal directions for the reversed sites.

et al. (1996) and Saddiqi et al. (1995), it is implicit that the northward component is of thermal origin. We have observed that there is a relation between the relative intensity of this component and the serpentinization of the peridotites. Furthermore, within an outcrop, samples with a northward component present higher intensity than samples in which only the ChRM was observed. These results are more likely interpreted as being due to the presence of a new magnetic phase (also magnetite but with a coarser grain size distribution) associated with the last serpentinization process and not a partial TRM. In addition, normal and reversed samples exhibit similar rotations (Fig. 13), and if a very fast rotation rate is expected, differences should be observed between the two groups of samples. Finally, if the peridotites from Ojén Massif were cooled in no more than 1 Ma, it would be very difficult to explain a rotation rate of more than 40° per million years in this massif.

7. Conclusions

This study has shown that a mixture of SD and MD magnetite dominates the magnetic remanence of the peridotites. The behaviour of the samples during NRM demagnetization leads to their classification in four distinct groups, according to the appearance of a northward component superimposed on a characteristic remanent magnetization component. In some cases, the northward component has unblocking temperatures up to 400 °C, in other cases, it dominates the whole NRM, and, yet, in many other cases, it does not exist at all. The presence of this northward component is related to MD magnetite originated during a late serpentinization stage associated to the extensional dismembering of the peridotite slab. The ChRM is related to the presence of SD or PSD magnetite and has been interpreted as a thermochemical remanent magnetization acquired during the post-metamorphic cooling of the peridotites, between 20 and 17–18 Ma ago. After the cooling of the peridotite massifs, the region experienced clockwise block rotations about vertical axes of $41° \pm 12°$. Normal and reversed polarities are recorded by the ChRM, indicating that the Earth's magnetic field experienced at least one reversal during ChRM acquisition. The polarity distribution of the ChRM points to the identification of a polarity zonation presumably related to the cooling process. The negative result of a fold test in which the tilt of the compositional layering has been used as a tectonic correction indicates that the compositional layering was not a palaeohorizontal at the time of ChRM acquisition.

Since rotations took place after the cooling of the massifs (i.e., after 17–18 Ma), the structures developed within the peridotites prior to their cooling are systematically rotated clockwise. This result must be

considered when modelling the origin of these peridotitic bodies.

Finally, geodynamic models proposed to explain the geological history of the Betic Cordilleras must consider the constraints imposed by these palaeomagnetic data. The models should explain that this region has experienced a clockwise rotation of about 40° after the cooling of the peridotites below, at least, 350–400 °C (i.e., after 20–17 Ma).

Acknowledgements

This work could not be developed without the invaluable help of: C. Osete and F. Martín-Hernández during the first campaign; C. Osete, J. Villalaín J.I. Núñez and A. Palencia during the second campaign. We would like to thank J. Gee and L. Tauxe for their support during the experiments performed in Scripps Palaeomagnetic Laboratory. We also want to thank the directors and staff of Zürich and Utrecht Palaeomagnetic Laboratories for their advice and the use of their equipment. Finally, we would like to thank the referees of this article, especially J.M. Tubía, whose criticism and suggestions have greatly improved the paper. This work has been supported by the Dirección General de Investigaciones Científicas y Tecnológicas (DGICYT, projects PB98-0834 and BTE2002-00854).

References

Allerton, S., 1994. Vertical axis rotation associated with folding and thrusting: an example from the eastern Subbetic of southern Spain. Geology 22, 1039–1042.

Allerton, S., Lonergan, L., Platt, J.P., Platzman, E., McClelland, E., 1993. Palaeomagnetic rotations in the eastern Betic Cordillera, southern Spain. Earth Planet. Sci. Lett. 119, 225–241.

Allerton, S., Reicherter, K., Platt, J.P., 1994. A structural and palaeomagnetic study of a section through the eastern Subbetic, Southern Spain. J. Geol. Soc. 151, 659–668.

Andrieux, J., Fontboté, J.M., Mattauer, M., 1971. Sur un modele explicatif de l'arc de Gibraltar. Bull. Sci. Geol. Fr. 7, 115–118.

Balanyá, J.C., 1991. Estructura del Dominio de Alborán en la parte norte del Arco de Gibraltar. PhD Thesis, Granada University.

Balanyá, J.C., García-Dueñas, V., 1987. Les directions structurales dans la domaine d'Alborán de part e d'autre du Detroit de Gibraltar. C. R. Acad. Sci., Paris 304, 929–933.

Barberá, X., Cabrera, L., Gomis, E., Parés, J.M., 1996. Determinación del polo paleomagnético para el límite Oligoceno–Mioceno en la Cuenca del Ebro. Geogaceta 20 (5), 1014–1016.

Bina, M.M., Henry, B., 1990. Magnetic properties, opaque mineralogy and magnetic anisotropies of serpentinized peridotites from ODP Hole 670[a] near the Mid-Atlantic Ridge. Phys. Earth Planet. Inter. 65, 88–103.

Calvo, M., Osete, M.L., Vegas, R., 1994. Palaeomagnetic rotations in opposite senses in southern Spain. Geophys. Res. Lett. 21, 761–764.

Calvo, M., Vegas, R., Osete, M.L., 1997. Palaeomagnetic results from Upper Miocene and Pliocene rocks from Internal Zone of the eastern Betic Cordilleras (southern Spain). Tectonophysics 227, 271–283.

Calvo, M.R., Cuevas, J., Tubía, J.M., 2001. Preliminary palaeomagnetic results on Oligocene–early Miocene mafic dykes from southern Spain. Tectonophysics 332, 333–345.

Cande, S.C., Kent, D.V., 1995. Revised calibration of the geomagnetic polarity timescale for the Late Cretaceous and Cenozoic. J. Geophys. Res. 100, 6093–6095.

Carey, S.W., 1955. The orocline concept in geotectonics. Proc. R. Soc. Tasmania 89, 255–288.

Chalouan, A., Michard, A., 1990. The Ghomarides nappes, Rif coastal range, Morocco: a Variscan chip in the Alpine belt. Tectonics 9, 1565–1583.

Day, R., Fuller, M., Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain-size and compositional dependence. Phys. Earth Planet. Inter. 13, 260–267.

Doblas, M., Oyarzun, R., 1989. Mantle complexes and Neogene extensional detachment tectonics in the western Betic Cordilleras, Spain: an alternative model for the emplacement of the Ronda peridotites. Earth Planet. Sci. Lett. 111, 11–21.

Docherty, J.I.C., Banda, E., 1995. Evidence for the eastward migration of the Alborán Sea based on regional subsidence analysis: a case for basin formation by delamination of the subcrustal lithosphere? Tectonics 14, 804–818.

Dunlop, D.J., 2002a. Theory and application of the Day plot (Mrs/Ms versus Hcr/Hc): 1. Theoretical curves and tests using titanomagnetite data. J. Geophys. Res. 107 (B3), EPM 4-1–EPM 4-22.

Dunlop, D.J., 2002b. Theory and application of the Day plot (Mrs/Ms versus Hcr/Hc): 2. Application to data for rocks, sediments and soils. J. Geophys. Res. 107 (B3), EPM 5-1–EPM 5-15.

Dunlop, D.J., Özdemir, Ö., 1997. Rock Magnetism: Fundamentals and Frontiers. Cambridge University Press, Cambridge, UK.

Dunlop, D.J., Prévot, M., 1982. Magnetic properties and opaque mineralogy of drilled submarine intrusive rocks. Geophys. J. R. Astron. Soc. 69, 763–802.

Elazzab, D., Feinberg, H., 1994. Paleomagnetisme des roches ultrabasiques du Rif externe (Maroc). C. R. Acad. Sci., Paris 318, 351–357.

Feinberg, H., Saddiqi, O., Michard, A., 1996. New constraints on the bending of the Gibraltar Arc from palaeomagnetism from the Ronda peridotites (Betic Cordilleras, Spain). In: Morris, A., Tarling, D.H. (Eds.), Palaeomagnetism and Tectonics of the Mediterranean Region. Geological Society of London, Special Publication, vol. 105, pp. 43–52.

García-Dueñas, V., Balanyá, J.C., Martínez-Martínez, J.M., 1992.

Miocene extensional detachments in the outcropping basement of the North Alborán Basin (Betics) and their tectonic implications. Geol. Marine Lett. 12, 88–95.

Gervilla, F., 1990. Mineralizaciones magmaticas ligadas a la evolucion de los macizos ultramaficos de la Serrania de Ronda (Malaga, España). PhD Thesis, Granada University.

Kirker, A., McClelland, E., 1996. Application of net tectonic rotations and inclination analysis to a high resolution palaeomagnetic study in the Betic Cordillera. In: Morris, A., Tarling, D.H. (Eds.), Palaeomagnetism and Tectonics of the Mediterranean Region. Geological Society of London, Special Publication, vol. 105, pp. 19–32.

Leblanc, D., Olivier, P., 1984. Role of strike-slip faults in the Betic–Riffean orogeny. Tectonophysics 101, 345–355.

Loomis, T.P., 1972. Diapiric emplacement of the Ronda high temperature ultramafic intrusion, southern Spain. Geol. Soc. Amer. Bull. 83, 2475–2496.

Loomis, T.P., 1975. Tertiary mantle diapirism, orogeny and plate tectonics east of the Strait of Gibraltar. Am. J. Sci. 275, 1–30.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. Geophys. Res. Lett. 17, 159–162.

Lundeen, M., 1978. Emplacement of the Ronda peridotite, Sierra Bermeja, Spain. Geol. Soc. Amer. Bull. 89, 172–180.

Michard, A., Chalouan, A., Montigny, R., Touhami Ouazzani, M., 1983. Les nappes crystallophylliennes du Rif (Sebtides, Maroc), temoins d'un edifice alpin de type pennique incluant le manteau superieur. C.R. Acad. Sci., Paris Ser. II 296, 1337–1340.

Moody, J.B., 1976. An experimental study on the serpentinization of iron-bearing olivines. Can. Mineral. 14, 462–478.

Obata, M., 1980. The Ronda peridotite: garnet-, spinel-, and plagioclase-lherzolite facies and the P–T trajectories of a high temperature mantle intrusion. J. Petrol. 21, 533–572.

Osete, M.L., Freeman, R., Vegas, R., 1988. Preliminary palaeomagnetic results from the Subbetic Zone (Betic Cordillera, southern Spain); kinematic and structural implications. Phys. Earth Planet. Inter. 52, 283–300.

Osete, M.L., Freeman, R., Vegas, R., 1989. Palaeomagnetic evidence for block rotations and distributed deformation of the Iberian–African plate boundary. In: Kissel, C., Laj, C. (Eds.), Palaeomagnetic Rotations and Continental Deformation. NATO ASI Series, Mathematical and Physical Sciences, vol. 254. Kluwer Academic Publishing, Dordrecht, pp. 381–391.

Osete, M.L., Villalaín, J.J., Palencia, A., Osete, C., Sandoval, J., García-Dueñas, V., 2004. New palaeomagnetic data from the Betic Cordillera. Constraints on the timing and the geographical distribution of tectonic rotations in southern Spain. Pure Appl. Geophys. (in press).

Platt, J.P., Vissers, R.L.M., 1989. Extensional collapse of thickened continental lithosphere: a working hypothesis for the Alborán Sea and Gibraltar Arc. Geology 17, 540–543.

Platt, J.P., Allerton, S., Kirker, A., Platzman, E., 1994. Origin of the western Subbetic (South Spain): palaeomagnetic and structural evidence. Geology 17, 765–775.

Platt, J.P., Soto, J.I., Whitehouse, M.J., Hurford, A.J., Kelley, S.P., 1998. Thermal evolution, rate of exhumation, and tectonic significance of metamorphic rocks from the floor of the Alboran extensional basin, western Mediterranean. Tectonics 17 (5), 671–689.

Platzman, E., 1992. Palaeomagnetic rotations and the kinamatics of the Gibraltar Arc. Geology 20, 311–314.

Platzman, E., Lowrie, W., 1992. Palaeomagnetic evidence for rotation of the Iberian Peninsula and the External Betic Cordillera, Southern Spain. Earth Planet. Sci. Lett. 108, 45–60.

Platzman, E., Platt, J.P., Olivier, P., 1993. Palaeomagnetic rotations and fault kinematics in the Rif arc of Morocco. J. Geol. Soc. (Lond.) 150, 707–718.

Platzman, E., Platt, J.P., Kelley, S.P., Allerton, S., 2000. Large clockwise rotations in an extensional allochthon, Alborán Domain (southern Spain). J. Geol. Soc. (Lond.) 157, 1187–1197.

Polvé, M., 1983. Les isotopes du Nd et du Sr dans les lherzolites orogeniques: contribution à la determination de la structure et de la dynamique du Manteau Superieur. Thèse Doctoral, Université de París, París, France.

Priem, H.N.A., Boelrijk, N.A.I.M., Hebeda, E.H., Oen, I.S., Verdurmen, E.A.T., Verschure, R.H., 1979. Isotopic dating of the emplacement of the ultramafic masses in the Serranía de Ronda, southern Spain. Contrib. Mineral. Petrol. 70, 103–109.

Reisberg, L., Zindler, A., Jagoutz, E., 1989. Further Sr and Nd isotopic results from peridotites of the Ronda Ultramafic Complex. Earth Planet. Sci. Lett. 96, 161–180.

Reisberg, L., Allegre, C.J., Luck, J.M., 1991. The Re–Os systematics of the Ronda Ultramafic Complex of southern Spain. Earth Planet. Sci. Lett. 105, 196–213.

Saddiqi, O., Feinberg, H., Elazzab, D., Michard, A., 1995. Paleomagnetisme des roches ultrabasiques des Beni Bousera, Rif septentrional, Maroc. Comptes Rendus de l'Academie des Sciences, Paris 321, 361–368.

Sánchez-Rodríguez, L., Gebauer, D., 2000. Mesozoic formation of pyroxenites and gabbros in the Ronda Area (southern Spain), followed by Early Miocene subduction metamorphism and emplacement into the middle crust: U–Pb sensitive high resolution ion microprobe dating of zircon. Tectonophysics 316, 19–44.

Seideman, D.E., 1976. An $^{40}Ar/^{39}Ar$ age spectrum for a cordierite bearing rock: isolating the effect of excess radiogenic ^{40}Ar. Earth Planet. Sci. Lett. 33, 268–272.

Torné, M., Banda, E., García-Dueñas, V., Balanyá, J.C., 1992. Mantle lithosphere bodies in the Alborán crustal domain (Ronda peridotites, Betic–Rif orogenic belt). Earth Planet. Sci. Lett. 110, 163–171.

Tubía, J.M., Cuevas, J., 1986. High-temperature emplacement of the Los Reales peridotite nappe (Betic Cordillera, Spain). J. Struct. Geol. 8, 473–482.

Tubía, J.M., Cuevas, J., 1987. Structures et cinematique liees a la mise en place des peridotites de Ronda (Cordilleres Betiques, Espagne). Geodyn. Acta 1, 59–69.

Van der Wal, D., Vissers, R.L.M., 1993. Uplift and emplacement of upper mantle rocks in the western Mediterranean. Geology 21, 1119–1122.

Vauchez, A., Nicolas, A., 1991. Mountain building: strike-parallel motion and mantle anisotropy. Tectonophysics 185, 183–201.

Villalaín, J.J., Osete, M.L., Vegas, R., García-Dueñas, V., Heller, F., 1994. Widespread Neogene remagnetization in Jurassic limestones of the South-Iberian palaeomargin (Western Betics, Gibraltar Arc). Phys. Earth Planet. Inter. 85, 15–33.

Villalaín, J.J., Osete, M.L., Vegas, R., García-Dueñas, V., Heller, F., 1996. The Neogene remagnetization in the western Betics: a brief comment on the reliability of palaeomagnetic directions. In: Morris, A., Tarling, D.H. (Eds.), Palaeomagnetism and Tectonics of the Mediterranean Region. Geological Society of London, Special Publication, vol. 105, pp. 33–41.

Weijermaars, R., 1985. Uplift and subsidence history of the Alborán Basin and the profile of the Alborán Diapir (W. Mediterranean). Geol. Mijnb. 64, 349–356.

Zindler, A., Staudiget, H., Hart, S.R., Endres, R., Goldstein, S., 1983. Nd and Sr isotopic study of a mafic layer from Ronda ultramafic complex. Nature 304, 226–230.